

Producto interno de Hilbert-Schmidt

Condiciones de un producto interno:

$$1: \langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$$

$$2: \langle \alpha \vec{v}, \vec{w} \rangle = \alpha \langle \vec{v}, \vec{w} \rangle$$

$$3: \langle \vec{v}, \vec{w} \rangle = \overline{\langle \vec{w}, \vec{v} \rangle} \rightarrow \text{Producto interno hermitiano}$$

$$4: \langle \vec{v}, \vec{v} \rangle \geq 0, \text{ igual si y solo si } \vec{v} = 0$$

Hilbert-Schmidt:

Es un producto interno para matrices complejas y hermiticas cuadradas.

$$\langle A, B \rangle = \text{Tr}(A^\dagger B)$$

Verificación de las condiciones

$$1: \text{Tr}[(A+B)^\dagger C] = \text{Tr}[(A^\dagger + B^\dagger)C] = \text{Tr}(A^\dagger C + B^\dagger C) = \text{Tr}(A^\dagger C) + \text{Tr}(B^\dagger C) \quad \checkmark$$

$$2: \text{Tr}[(\alpha A)^\dagger B] = \alpha^* \text{Tr}(A^\dagger B)$$

$$3: \text{Tr}(A^\dagger B)^* = \text{Tr}(A^\dagger B)^\dagger = \text{Tr}(B^\dagger A) \quad \checkmark$$

$$4: \text{Tr}(A^\dagger A) = a^*a + b^*b + c^*c + d^*d + \dots \geq 0 \quad \checkmark$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} \Rightarrow A^\dagger A = \begin{pmatrix} a^*a + c^*c & a^*b + c^*d \\ b^*a + d^*c & b^*b + d^*d \end{pmatrix}$$

$$(x+iy)(x-iy) = x^2 + y^2 \geq 0$$